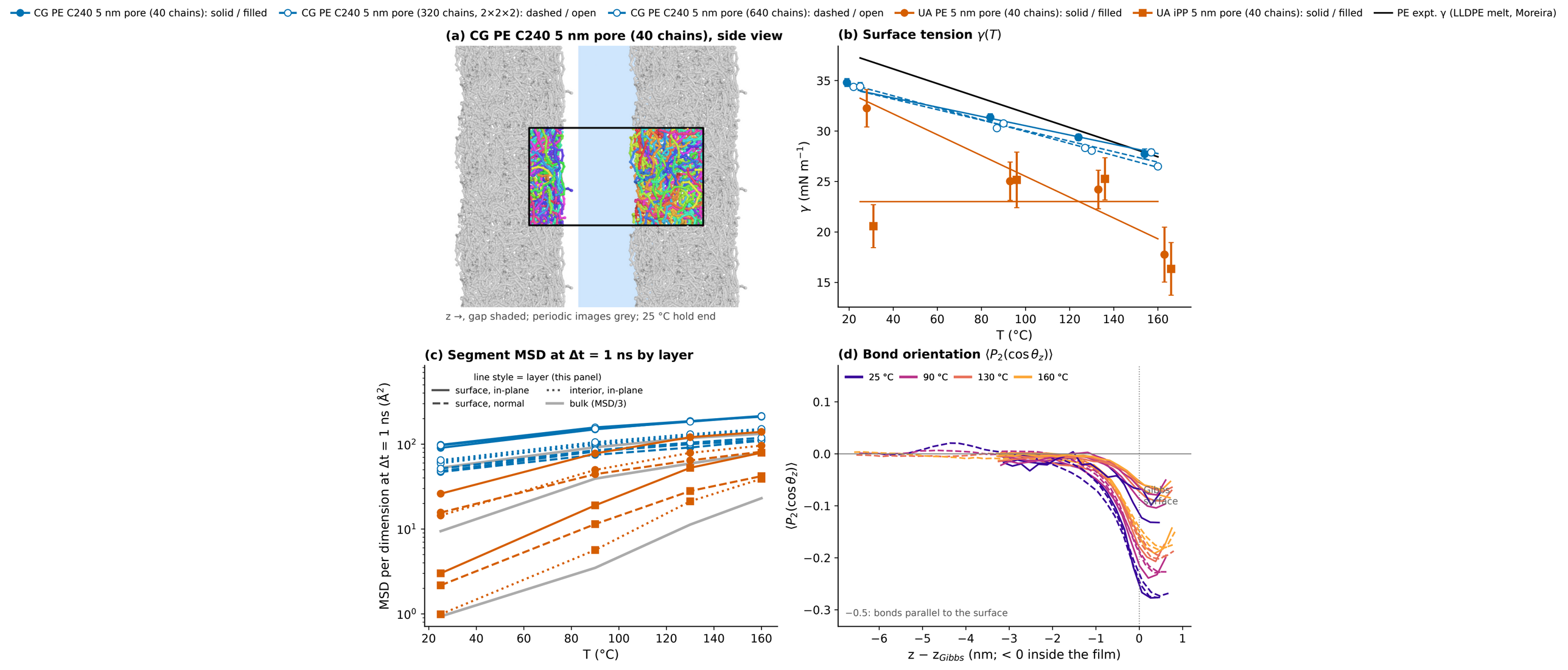

CG PE C240 5 nm pore (40 chains): solid / filled
CG PE C240 5 nm pore (320 chains, 2×2×2): dashed / open
CG PE C240 5 nm pore (640 chains): dashed / open
UA PE 5 nm pore (40 chains): solid / filled
UA iPP 5 nm pore (40 chains): solid / filled
PE expt. γ (LLDPE melt, Moreira)
(a) CG PE C240 5 nm pore (40 chains), side view
z →, gap shaded; periodic images grey; 25 °C hold end
(b) Surface tension γ(T)
γ (mN m−1)
T (°C)
(c) Segment MSD at Δt = 1 ns by layer
line style = layer (this panel)
surface, in-plane
interior, in-plane
surface, normal
bulk (MSD/3)
MSD per dimension at Δt = 1 ns (Å2)
T (°C)
(d) Bond orientation ⟨P2(cos θz)⟩
25 °C
90 °C
130 °C
160 °C
Gibbs surface
−0.5: bonds parallel to the surface
⟨P2(cos θz)⟩
z − zGibbs (nm; < 0 inside the film)